From Prediction to Design: A Paradigm Shift in AI-Driven Protein-Protein Interaction Research

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Abstract

Protein-protein interactions (PPIs) form the molecular foundation of cellular life. Over the past two decades, computational methods for PPI analysis have primarily focused on *prediction*—inferring interaction properties from protein information. However, a fundamental paradigm shift is underway: from "predicting what will happen" to "designing what we want." This transformation is driven by breakthroughs in protein language models, diffusion models, and equivariant neural networks. In this review, we argue that this transition constitutes a genuine paradigm shift in the Kuhnian sense, enabling investigation of sequence-structure space that evolution never explored. We systematically analyze the technical foundations, core design tasks, and challenges ahead. We address critical issues including evaluation bias, the hallucination problem, and dual-use considerations. We conclude with grand challenges for the field: mapping designability boundaries, designing dynamic interactions, and developing interpretable methods for clinical translation. This paradigm shift marks a fundamental change in how AI interfaces with biological discovery—from passive observation to active creation.

Keywords: protein-protein interactions, deep learning, generative models, protein design, AlphaFold, diffusion models, protein language models, paradigm shift

Contents

1 Introduction **3**

2 The Paradigm Transition: From Prediction to Design **4**
2.1 The Prediction Paradigm: Achievements and Limitations 5
2.2 Technical Drivers of the Design Paradigm 6
2.3 Milestone Works in the Design Era 6
2.4 The Complementarity of Prediction and Design 7
2.5 Critical Assessment: The Hallucination Problem 7

3 Core Design Tasks **8**
3.1 Protein Binder Design 8
3.2 Antibody Design and Optimization 8
3.3 Interface Optimization 8
3.4 De Novo Interaction Networks 9

4 Enabling Technologies **9**
4.1 Generative Models for Protein Design 9
4.2 Structure-Aware Representations 10
4.3 Protein Language Models 11

5 Challenges and Limitations **11**
5.1 Evaluation and Benchmarking . . . 11
5.2 Experimental Validation Bottleneck . . . 12
5.3 Exploring the Designable Space . . . 13
5.4 Dynamic and Conditional Interactions . . . 13
5.5 Current Approaches to Dynamic PPIs . . . 13
5.6 Interpretability and Trust . . . 14
5.7 Ethical and Dual-Use Considerations . . . 14

6 Applications and Impact **14**
6.1 Therapeutic Protein Design . . . 14
6.2 Industrial and Economic Impact . . . 15
6.3 Research Tools and Basic Science . . . 16
6.4 Economic and Accessibility Considerations . . . 16

7 Future Perspectives **16**
7.1 Prediction-Design Integration . . . 17
7.2 Toward Protein Intelligence . . . 17
7.3 Methodological Advances . . . 17
7.4 Experimental Infrastructure . . . 17
7.5 Interpretability and Clinical Translation . . . 17
7.6 Paradigm Shifts in Research Practice . . . 18
7.7 Accessibility and Equity . . . 18
7.8 Grand Challenges for the Next Five Years . . . 18

8 Conclusion **18**

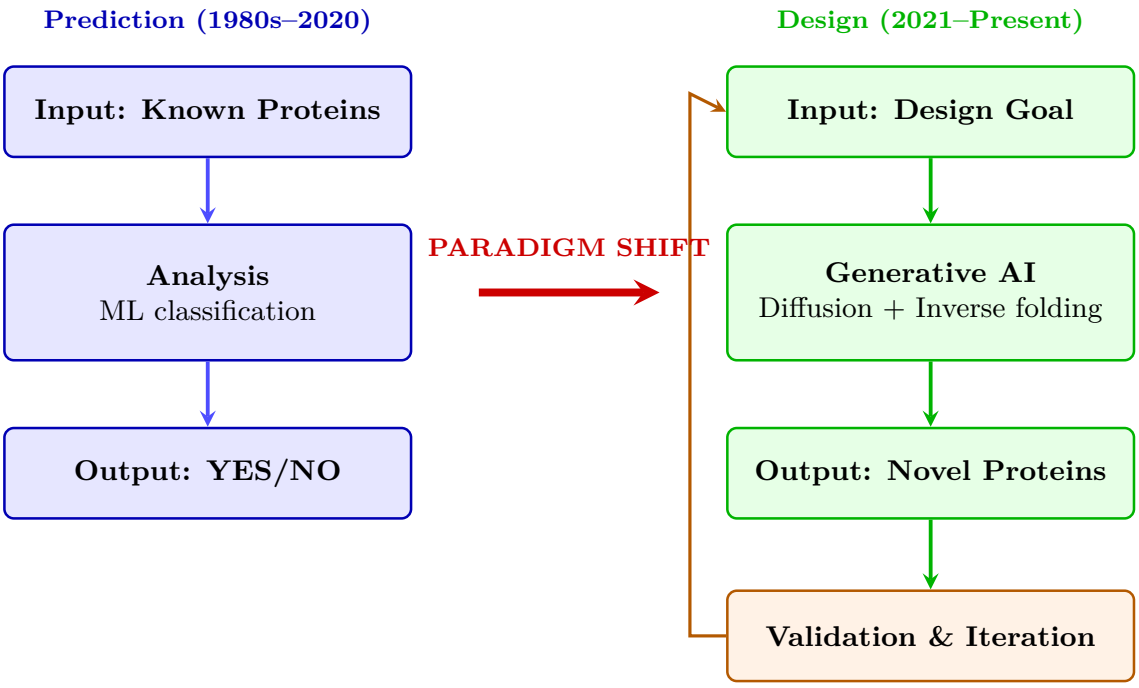

Figure 1: The paradigm shift from prediction to design in AI-driven PPI research. **Left panel (Prediction Paradigm, 1980s–2020):** Traditional computational approaches analyze known protein sequences and structures to predict interaction likelihood (YES/NO). Blue color scheme indicates analytical methods including machine learning classification and structure comparison. **Right panel (Design Paradigm, 2021–present):** Modern AI-driven approaches generate novel protein sequences with desired interaction properties through iterative design-test cycles. Green color scheme indicates generative methods including diffusion models and inverse folding. The validation loop (orange) reflects the critical role of experimental feedback.

1 Introduction

Protein-protein interactions (PPIs) form the molecular foundation of cellular life. The human interactome is estimated to contain 130,000–650,000 binary interactions Szklarczyk et al. [2019], with disease-associated mutations disproportionately affecting interaction interfaces Sahni et al. [2015]. Over the past two decades, computational approaches to PPI research have primarily focused on *prediction*: given protein sequences or structures, infer whether and how they interact. This prediction paradigm has produced remarkable successes, including high-accuracy interaction prediction from sequence alone Chen et al. [2019a], Sledzieski et al. [2021] and unprecedented accuracy in complex structure prediction Evans et al. [2022], Abramson et al. [2024].

Yet a fundamental limitation constrains prediction methods: they can only describe what might happen, not create what we want to happen. For drug discovery and synthetic biology, the relevant question is not "will these proteins interact?" but "how can we design proteins that interact the way we need?" This shift from passive observation to active creation defines the emerging *design paradigm*.

Defining the Paradigm Shift

We argue that the transition from prediction to design represents a qualitative transformation in computational PPI research. The prediction paradigm asks: "What interactions exist in nature?" The design paradigm asks: "What interactions can we create?"

This shift is characterized by three features. **First**, the design paradigm enables investigation

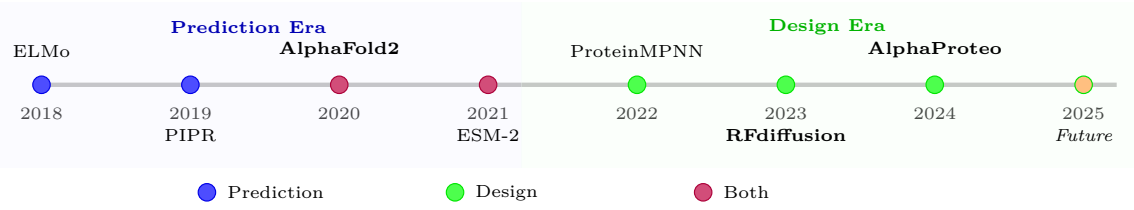

Figure 2: Timeline of major technical breakthroughs in AI-driven protein research (2018–2025). The paradigm shift from prediction to design occurred around 2021–2022, marked by the emergence of generative methods. **Milestones in bold** (AlphaFold2, RFdiffusion, AlphaProteo) represent transformative advances with extensive experimental validation. **Color coding:** Blue = prediction-focused; green = design-focused; purple = spanning both paradigms. Shaded regions delineate the Prediction Era (2018–2021) from the Design Era (2022–present).

of previously inaccessible regions of sequence-structure space—configurations that evolution never explored but that may contain superior therapeutic or industrial properties. **Second**, the design paradigm requires fundamentally different methodological approaches: generative models rather than discriminative classifiers, inverse problems rather than forward prediction, and iterative design-test cycles rather than one-shot inference. **Third**, prediction and design are complementary rather than incommensurable: every design system relies on prediction capabilities for validation Watson et al. [2023], and prediction methods benefit from novel test cases that design provides. This complementarity distinguishes our view from strong Kuhnian incommensurability Kuhn [1962] while preserving the paradigm shift characterization.

Technological Foundations and Controversies

This paradigm shift is enabled by converging technological advances: protein language models encoding evolutionary information at scale Rives et al. [2021], Lin et al. [2023], diffusion models generating novel structures conditioned on functional constraints Watson et al. [2023], Ingraham et al. [2023], and equivariant neural networks preserving geometric symmetries Jing et al. [2021], Fuchs et al. [2020].

The field is not without controversy. Critics note that success rates for de novo design remain modest (10–50% depending on target) compared to natural protein engineering Tsishyn and Rooman [2024]. Questions persist about whether designed proteins will exhibit the stability, specificity, and lack of immunogenicity required for therapeutic use. A critical evaluation crisis has been identified: Tsishyn and Rooman Tsishyn and Rooman [2024] demonstrated that many reported high accuracies result from dataset-specific biases rather than genuine predictive capability, suggesting that methodological artifacts may inflate perceived progress.

Article Structure

This review provides a comprehensive analysis of this paradigm transition. Section 2 examines the trajectory from prediction to design, including methodological evolution and key milestones. Section 3 surveys core design tasks enabled by new capabilities. Section 4 details enabling technologies. Section 5 discusses open challenges including evaluation, validation, and interpretability. Section 6 examines therapeutic and industrial applications. Section 7 offers perspectives on future directions, and Section 8 synthesizes key themes and outlines grand challenges for the field.

2 The Paradigm Transition: From Prediction to Design

The field of computational PPI research has evolved through distinct methodological eras, culminating in the current design-oriented approaches. This section traces this evolution, examining how prediction capabilities laid the foundation for design.

2.1 The Prediction Paradigm: Achievements and Limitations

Before the emergence of generative design, PPI research was dominated by the prediction paradigm—inferring whether, where, and how strongly proteins interact based on their sequences or structures.

Sequence-Based Prediction. Methods evolved through three generations: (1) handcrafted feature approaches using SVMs with physicochemical properties Bock and Gough [2001], Shen et al. [2007], Ben-Hur and Noble [2005]; auto-covariance descriptors captured interactions between residues at varying sequence distances Guo et al. [2008], You et al. [2017]; (2) deep learning methods enabling automatic feature learning Chen et al. [2017], Hashemifar et al. [2018], Chen et al. [2019a]; and (3) pre-trained protein language models leveraging evolutionary information Rives et al. [2021], Sledzieski et al. [2021]. Chinese research groups contributed substantially to sequence-based PPI prediction Lv et al. [2021], Tang et al. [2021]. Graph neural networks emerged as powerful approaches, with DeepRank-GNN Bryant et al. [2022a] learning patterns directly from structural graphs. TUnA introduced uncertainty quantification Gehring et al. [2024], while TAGPPI incorporated predicted structural information Xie et al. [2022]. D-SCRIPT demonstrated that structure-aware representations improve cross-species generalizability Sledzieski et al. [2021], and Topsy-Turvy Singh et al. [2022] integrated sequence-based predictions with network topology.

Structure-Based Prediction. Classical protein-protein docking methods predict bound complex structures from individual proteins. HADDOCK Dominguez et al. [2003] introduced information-driven docking with biochemical constraints, ZDOCK Pierce et al. [2014] uses fast Fourier transforms for shape complementarity, ClusPro Kozakov et al. [2017] provides automated docking through clustering, HDOCK Yan et al. [2017] combines template-based and free docking, and GalaxyTongDock Baek et al. [2017] enables symmetric and asymmetric docking with improved energy parameters. Surface-based deep learning methods predict binding interfaces directly from molecular structures: MaSIF Gainza et al. [2020] applied geometric deep learning to surfaces, GraphPPIS Cao et al. [2022] used graph convolutional networks, dMaSIF Sverrisson et al. [2021] achieved 10–100× speedups with point-cloud operators, and ScanNet Tubiana et al. [2022] combined geometric deep learning with spatial attention for interpretable binding site prediction. Beyond binary interaction prediction, binding site prediction identifies which residues mediate interactions—critical information for drug design and mutagenesis studies García-Ortegón et al. [2022]. Systematic benchmarking of AlphaFold for protein complexes revealed important determinants of prediction accuracy Bryant et al. [2022b]. The most dramatic advance is end-to-end complex prediction: AlphaFold-Multimer Evans et al. [2022] and AlphaFold 3 Abramson et al. [2024] predict complex structures from sequences alone with unprecedented accuracy.

Table 1: Taxonomy of PPI prediction tasks

Task	Input	Output	Representative Methods
Binary classification	Sequence pairs	Interaction (yes/no)	PIPR Chen et al. [2019a], D-SCRIPT Sledzieski et al. [2021]
Network inference	Protein set	Interaction network	STRING Szklarczyk et al. [2019], GNN-PPI Lv et al. [2021]
Binding site prediction	Single protein	Interface residues	MaSIF Gainza et al. [2020], GraphPPIS Cao et al. [2022]
Affinity prediction	Complex + mutation	$\Delta\Delta G$	FoldX Guerois et al. [2003], SKEMPI Jankauskaite et al. [2019]
Complex structure	Sequence pairs	3D coordinates	AlphaFold-Multimer Evans et al. [2022], DiffDock-PP Ketata et al. [2023]

Fundamental Limitations. Despite progress, the prediction paradigm faces inherent

constraints:

1. *Data Dependence*: Models learn statistical patterns from known interactions and struggle to extrapolate to novel types Kewalramani and Emili [2023]. Training data is heavily biased toward model organisms (human, yeast, mouse). Negative sample construction introduces systematic bias—randomly paired proteins may not represent true non-interactors Smialowski et al. [2010]. The Negatome database addressed this by curating experimentally verified non-interacting pairs Blohm et al. [2014]. Prediction accuracy degrades significantly for proteins dissimilar from training examples, as demonstrated by the C1/C2/C3 difficulty classification framework Park and Marcotte [2015].

2. *Evaluation Crisis*: A systematic analysis of binding affinity prediction methods by Tsishyn and Rooman Tsishyn and Rooman [2024] revealed that many reported accuracies are inflated by dataset-specific biases rather than genuine predictive capability. Eight widely-used predictors showed strong dependence on training datasets, with performance degrading substantially when evaluated on proteins from different families. Benchmark datasets suffer from incompleteness and bias, with STRING Szklarczyk et al. [2019], BioPlex Huttlin et al. [2021], and IntAct Orchard et al. [2014] each having different coverage and confidence scoring schemes Feng et al. [2022]. This suggests that current train-test splitting protocols may be fundamentally inadequate for evaluating generalization to novel protein interactions.

3. *Passive Discovery Mode*: Prediction methods answer "will these proteins interact?" rather than "how can we make them interact?" For therapeutic applications, this provides no direct pathway from prediction to intervention.

4. *Static Assumptions*: Most methods treat interactions as binary states, ignoring tissue-specificity, condition-dependence, and conformational dynamics McGary et al. [2010].

These limitations motivated the emergence of design-oriented approaches.

2.2 Technical Drivers of the Design Paradigm

The transition to design was enabled by converging breakthroughs occurring between 2021 and 2024.

Structure Prediction Advances. AlphaFold2 Jumper et al. [2021] and RoseTTAFold Baek et al. [2021] achieved near-experimental accuracy in structure prediction. Extensions to multimeric complexes Evans et al. [2022], Baek and Baker [2023] expanded scope to PPIs. These advances provide both validation mechanisms for designed proteins and learned representations repurposed for generation.

Generative Models. Diffusion models achieved breakthrough performance by treating protein generation as learning to reverse a noise-adding process Ho et al. [2020]. RFdiffusion Watson et al. [2023] fine-tuned RoseTTAFold as a denoising model, enabling conditional generation on partial structures or target proteins. Chroma Ingraham et al. [2023] incorporated physical constraints directly into the generative process, enabling generation of proteins up to 4,000 amino acids. Flow-based alternatives like FoldFlow Bose et al. [2023] offer potentially better computational efficiency.

Protein Language Models at Scale. ESM-2 Lin et al. [2023], scaled to 15 billion parameters, achieved atomic-level structure prediction directly from sequence. These models provide learned priors over protein sequences—encoding what sequences are likely to fold into stable, functional proteins. Recent extensions like MINT Ullanat et al. [2025] and PLM-interact Zhao et al. [2025] train specifically on PPI datasets.

Equivariant Neural Networks. SE(3)-Transformers Fuchs et al. [2020] and Geometric Vector Perceptrons Jing et al. [2021] process geometric data while respecting rotational and

translational symmetries—critical for ensuring that generated proteins' validity does not depend on arbitrary coordinate systems. AlphaFold2's Invariant Point Attention Jumper et al. [2021] provided an influential alternative approach.

2.3 Milestone Works in the Design Era

The convergence of these technologies produced breakthrough systems organized by primary function (Table 2).

Table 2: Comparison of major AI-driven protein design tools (2022–2024)

Method	Year	Input/Output	Performance	Validation
RFdiffusion Watson et al. [2023]	2023	Target → Binder	10–50% binding[†]	Cryo-EM (100s)
AlphaProteo Zambaldi et al. [2024]	2024	Target → Binder	3–300× vs baseline[‡]	SPR/BLI (nM)
ProteinMPNN Dauparas et al. [2022]	2022	Backbone → Sequence	52.4% native recovery[§]	X-ray validation
ESM-IF1 Hsu et al. [2022]	2022	Backbone → Sequence	51% native recovery[§]	12M structures
Chroma Ingraham et al. [2023]	2023	Constraints → Full protein	70–90% designability[‖]	MD simulation
DiffDock-PP Ketata et al. [2023]	2023	Structures → Docked poses	44% top-10 pose[¶]	Benchmark

[†]Binding success rate varies by target difficulty; data from Watson et al. Watson et al. [2023].
[‡]Improvement over RFdiffusion baseline on 7 targets; KD: 1–100 nM Zambaldi et al. [2024].
[§]Native sequence recovery on CATH 4.2 test set Dauparas et al. [2022], Hsu et al. [2022].
[‖]Designability: fraction with pLDDT > 80; 70–90% depending on length Ingraham et al. [2023].
[¶]Top-10 success on DBD5 benchmark with pose RMSD < 2Å Ketata et al. [2023].

Backbone Generation. RFdiffusion demonstrated successful binder design with experimental validation: cryo-EM structures confirmed atomic-level agreement with computational models (backbone RMSD: 0.5–1.5 Å). Chroma enabled generation of large proteins with explicit physical constraints.

Sequence Design. ProteinMPNN and ESM-IF1 address the inverse folding problem—given a backbone, predicting sequences that fold into it. Both achieve 50% native sequence recovery, enabling high-throughput sequence design for generated backbones.

Binder Design. AlphaProteo represents the most advanced demonstration to date, achieving binding success rates 3–300× higher than RFdiffusion-based approaches across seven diverse targets, with multiple binders achieving nanomolar affinities without experimental optimization.

Multi-Modal Systems. AlphaFold 3 Abramson et al. [2024] and RoseTTAFold All-Atom Krishna et al. [2024] extend to complexes containing proteins, nucleic acids, and small molecules, with RFdiffusionAA enabling de novo design of protein binding pockets around cofactors.

2.4 The Complementarity of Prediction and Design

Prediction and design are not opposing paradigms but complementary capabilities. Every design system relies on prediction: RFdiffusion was initialized from RoseTTAFold's pretrained weights; ProteinMPNN is trained to predict foldable sequences; AlphaProteo uses structure prediction for validation. The design process can be viewed as iterative prediction: structure prediction of targets, interface prediction for binding sites, backbone generation as inverse structure prediction, and complex prediction for validation.

However, the complementarity does not negate the paradigm shift characterization. Design enables qualitatively new capabilities—exploring sequence-structure space that evolution never accessed—while prediction provides the foundation upon which design builds. The relationship is bidirectional: design also advances prediction by providing test cases that reveal limitations in current understanding.

2.5 Critical Assessment: The Hallucination Problem

A fundamental challenge in generative protein design is distinguishing genuine design from "hallucination"—structures that appear protein-like to computational metrics but lack physical plausibility Anishchenko et al. [2021]. This manifests as backbone incoherence (violating Ramachandran constraints), sequence-structure mismatch (generated sequences not folding into designed structures), and energy landscape pathology (high-energy barriers preventing folding).

Current approaches mitigate hallucination through iterative validation: structure prediction on generated sequences, molecular dynamics simulation, and experimental characterization. However, this underscores that computational metrics cannot fully substitute for physical reality. A systematic study by Chu et al. Chu et al. [2025] found that roughly half of designed proteins failed at various experimental stages, with many failed designs having better computational confidence metrics than successful designs, highlighting the limitations of current predictive metrics.

Even experimental validation by cryo-EM and X-ray crystallography has caveats: these methods capture static snapshots that may not reflect dynamic behavior in solution. Comprehensive validation requires multiple complementary approaches: biophysical characterization for affinity, solution scattering for oligomeric state, and NMR for dynamics.

The paradigm shift is therefore not a complete replacement of traditional methods but a fundamental expansion of what is tractable. Design targets previously infeasible—novel scaffolds, de novo interfaces, challenging targets—are now accessible, while established methods excel in their domains of strength.

3 Core Design Tasks

The design paradigm enables fundamentally new questions. Rather than asking "what interactions exist?", researchers can now ask "how can we create interactions with desired properties?" This section surveys major design tasks enabled by recent advances.

3.1 Protein Binder Design

Protein binder design—creating novel proteins that specifically bind target molecules—represents the most direct application of the design paradigm Watson et al. [2023], Zambaldi et al. [2024]. Given a target structure, the goal is to generate a binding partner with high affinity and specificity.

The problem decomposes into subtasks: epitope selection (identifying the target region), scaffold generation (creating complementary backbone), sequence design (optimizing for stability and binding), and validation (predicting binding success). The challenge is that the design space is astronomically large—20^{100} possible sequences for a 100-residue protein—yet only a tiny fraction fold stably and bind with high affinity.

RFdiffusion Watson et al. [2023] pioneered general-purpose binder design. Given a target structure, it generates complementary backbones via diffusion, which ProteinMPNN Dauparas et al. [2022] then designs sequences for. Experimental validation demonstrated 10–50% success rates across diverse targets. AlphaProteo Zambaldi et al. [2024] integrated generation and validation into a unified pipeline, achieving 3–300× better success rates with nanomolar affinities on multiple therapeutic targets including PD-L1 and VEGF-A. For SARS-CoV-2, designed minibinders achieved picomolar affinities (KD: 10–100 pM), demonstrating that designed proteins can compete with evolved antibodies Watson et al. [2023].

3.2 Antibody Design and Optimization

Antibodies present unique challenges due to their conserved framework regions and hypervariable complementarity-determining regions (CDRs). Unlike other CDRs which adopt canonical structures encoded by germline genes, CDR-H3 lacks reliable structural templates Alford et al. [2017]. Its backbone conformation depends on the specific V(D)J recombination event, making prediction essentially a de novo folding problem. CDR-H3 length varies from 8 to 30+ residues (with longer loops increasingly common in therapeutic antibodies), and longer loops exhibit greater backbone conformational flexibility. Deep learning methods have made significant progress in antibody structure prediction Ruffolo et al. [2022, 2023], Henry et al. [2024], Abanades et al. [2023], and recent advances have improved antibody-antigen complex modeling Kenlay et al. [2024], Huang et al. [2024], but CDR-H3 modeling remains the primary bottleneck.

Antibody design involves: (1) designing binders against target antigens; (2) humanizing non-human antibodies to reduce immunogenicity; and (3) optimizing developability properties (stability, solubility, expression) Raybould et al. [2019]. Recent methods combine structural modeling with developability prediction to screen candidates before experimental validation. AbAgIntPre Li et al. [2022a] introduced sequence-only interaction prediction, while recent work applied prompt-based language models to nanobody-antigen interactions Zhang et al. [2024].

3.3 Interface Optimization

Rather than designing entirely new binding partners, interface optimization aims to improve existing interactions through targeted mutations Jankauskaite et al. [2019].

Affinity Maturation. Increasing binding strength is central to therapeutic optimization. Physics-based methods (FoldX Guerois et al. [2003], Rosetta Flex ddG Barlow et al. [2018]) estimate affinity changes from structural calculations but require accurate structures. Machine learning methods (mCSM-PPI2 Rodrigues et al. [2019], GeoPPI Liu et al. [2021]) learn statistical patterns from experimental data, achieving faster predictions. BeAtMuSiC Dehouck et al. [2013] introduced coarse-grained statistical potentials for rapid screening. SSIPe Li et al. [2022b] combines structural interface analysis with evolutionary information. DDAffinity Luo et al. [2024] uses geometric deep learning with attention for multiple simultaneous mutations, while recent work applied deep geometric representations to modeling mutation effects Qiu et al. [2024]. SSIF-Affinity Liu et al. [2024] integrates sequence, structure, and interaction features through multimodal learning.

Specificity Engineering. Beyond affinity, specificity—binding the intended target without binding similar off-targets—is critical for therapeutic applications. Designing proteins that discriminate between closely related paralogues requires subtle manipulation of interface chemistry, addressed through multi-objective optimization Rodrigues et al. [2019].

Stability-Developability Trade-offs. Interface mutations that improve affinity can destabilize the overall fold. Methods like ProteinMPNN explicitly optimize for predicted stability during sequence design Dauparas et al. [2022], while developability optimization addresses aggregation propensity, immunogenicity, and expression yield Raybould et al. [2019].

3.4 De Novo Interaction Networks

The most ambitious design task involves creating entire interaction networks—multiple proteins designed to interact in specific patterns Watson et al. [2023]. This enables synthetic biology applications from engineered signaling pathways to self-assembling biomaterials.

Symmetric oligomer design requires simultaneously optimizing interface geometry and overall assembly structure. RFdiffusion demonstrated successful design of cyclic oligomers Watson et al. [2023], with extensions enabling icosahedral cages Bennett et al. [2024]. Multi-component

systems present additional challenges: each protein must fold independently while forming specific interactions with partners. Current approaches decompose the problem into pairwise interfaces, but this cannot capture global system constraints.

Three fundamental challenges remain: orthogonality (ensuring proteins interact only with intended partners), kinetics (ensuring correct assembly order), and integration (functioning within cells without disrupting essential processes) Bennett et al. [2024]. These challenges are largely unaddressed by current methods.

These tasks share a common theme: they require generative capabilities beyond prediction. The enabling technologies underlying these capabilities are detailed in Section 4.

4 Enabling Technologies

The paradigm shift is enabled by three converging technological advances: generative models, structure-aware representations, and protein language models.

4.1 Generative Models for Protein Design

Generative models learn to sample from the distribution of valid protein structures, enabling creation rather than prediction. The key insight is that protein foldability and stability can be encoded in a learned energy landscape—the model implicitly captures biophysical constraints distinguishing natural proteins from random sequences Ho et al. [2020].

Diffusion Models. The dominant paradigm treats protein structure generation as learning to reverse a noise-adding process Ho et al. [2020]. The forward diffusion process gradually adds Gaussian noise:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}$$

where $\mathbf{x}_0$ is the original structure, $\mathbf{x}_t$ is the noisy version at timestep t, and $\bar{\alpha}_t$ is the noise schedule. The reverse process learns to denoise by predicting the added noise:

$$\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right]$$

RFdiffusion fine-tuned RoseTTAFold as a denoiser, exploiting transfer learning from pre-trained representations Watson et al. [2023]. Chroma incorporated physical constraints (bond lengths, angles, dihedral preferences) directly into the generative process Ingraham et al. [2023]. SE(3) diffusion models operate on the manifold of rigid body frames, where each residue is represented by position $\mathbf{r} \in \mathbb{R}^3$ and orientation $\mathbf{R} \in SO(3)$ Yim et al. [2023].

Flow-Based Alternatives. Flow matching learns deterministic trajectories through structure space rather than stochastic denoising. FoldFlow demonstrated comparable quality to diffusion with potentially better computational efficiency Bose et al. [2023], though diffusion currently leads in practical applications.

Inverse Folding. ProteinMPNN and ESM-IF1 address the complementary problem: given a backbone, what sequence will adopt it? Dauparas et al. [2022], Hsu et al. [2022]. These methods enable sequence design for generated backbones, completing the design pipeline.

4.2 Structure-Aware Representations

Effective protein modeling requires representations capturing both discrete chemical structure (sequence) and continuous geometry (3D coordinates). Protein function depends on 3D geometry, which must be represented in a coordinate-independent manner Fuchs et al. [2020], Jing et al. [2021].

Equivariant Neural Networks. SE(3)-Transformers Fuchs et al. [2020] and Geometric Vector Perceptrons (GVPs) Jing et al. [2021] process geometric data while respecting rotational

and translational symmetries. SE(3) equivariance requires that for any rotation $\mathbf{R} \in SO(3)$ and translation $\mathbf{t} \in \mathbb{R}^3$:

$$f(\mathbf{R}\mathbf{x} + \mathbf{t}) = \mathbf{R} \cdot f(\mathbf{x}) + \mathbf{t}$$

This ensures the same structure yields identical predictions regardless of orientation. GVPs achieve this through scalar-vector separation: each node maintains scalar features $\mathbf{s}$ and vector features $\mathbf{v}$, with message passing operating equivariantly on vectors while aggregating scalars invariantly. AlphaFold2's Invariant Point Attention (IPA) provided an alternative through learned reference frames Jumper et al. [2021], projecting 3D coordinates onto local frames where attention becomes rotation-invariant. IPA has become a foundational architectural element in subsequent design systems.

Why Equivariance Matters for Protein Design. The equivariance constraint is not merely a mathematical convenience—it reflects fundamental physical principles. Protein structures exist in 3D space independent of coordinate system orientation. A design method producing different outputs for rotated inputs violates this physical invariance. Critically, non-equivariant methods that work well for small molecules often fail for proteins: small molecule conformations can be enumerated densely, but protein conformational space is astronomically larger ($\sim 10^{130}$ for a 100-residue protein). Equivariance provides strong inductive bias that dramatically reduces effective search space. AlphaFold2's success relied heavily on this constraint: by ensuring equivariance, the model learns identical predictions regardless of input orientation, effectively multiplying training data by the symmetry group. The practical consequence for design is substantial: equivariant models require fewer training examples to achieve equivalent performance, critical given limited experimental validation data for designed proteins.

SE(3) Diffusion vs. Euclidean Diffusion. Standard diffusion operates in Euclidean space, treating coordinates independently. However, protein backbones lie on the SE(3) manifold—each residue has position $\mathbf{r} \in \mathbb{R}^3$ and orientation $\mathbf{R} \in SO(3)$. SE(3) diffusion respects this structure, adding noise through geodesic random walks rather than independent Gaussian noise. This ensures the noising process stays on the manifold of physically valid protein structures.

Geometric Deep Learning. MaSIF and PeSTo learn from molecular surfaces without relying on sequence information, demonstrating that interaction propensities are encoded in surface geometry and charge distribution Gainza et al. [2020], Rodríguez et al. [2023]. dMaSIF replaced expensive mesh generation with point-cloud operators, achieving 10–100× speedups Sverrisson et al. [2021]. The biological implication is that physical determinants of protein interactions—shape complementarity, electrostatic compatibility, hydrogen bonding—can be learned from structural data and transferred to design novel interfaces.

4.3 Protein Language Models

Protein language models learn representations from evolutionary information encoded in sequence databases, providing complementary structural and functional knowledge Rives et al. [2021], Lin et al. [2023].

Foundation Models. ESM-1b demonstrated that large-scale pre-training on protein sequences—without explicit structural supervision—yields representations capturing folding patterns, functional sites, and evolutionary constraints Rives et al. [2021]. ESM-2 scaled to 15 billion parameters, achieving atomic-level structure prediction from single sequences and eliminating the need for multiple sequence alignments Lin et al. [2023]. ESMFold runs 60× faster than AlphaFold2 while maintaining competitive accuracy. ProtTrans Elnaggar et al. [2021] established that encoder-only models outperform encoder-decoder architectures for protein understanding tasks. xTrimoPGLM scaled to 100 billion parameters for unified pre-training, representing a significant contribution from Chinese research groups to the foundation model landscape Chen et al. [2024]. GOProteinGNN Zhao et al. [2024] integrated protein knowledge graphs with graph neural networks leveraging Gene Ontology annotations.

Structure-Aware Language Models. SaProt introduced structure-aware tokenization where each token represents both sequence position and local structural state Su et al. [2024]. MINT and PLM-interact extended PLMs for interaction modeling, operating on protein sets to capture interaction propensities contextually Ullanat et al. [2025], Zhao et al. [2025].

Transfer Learning. The scale of PLMs enables effective fine-tuning for downstream tasks including binding site prediction, function annotation, and interaction prediction Dallago et al. [2024]. This reduces data requirements for specialized tasks while leveraging knowledge encoded in foundation models.

Computational Considerations. Training large PLMs requires enormous investment (ESM-2 required hundreds of GPU-years). Inference costs vary dramatically: ProteinMPNN runs on CPU in milliseconds, while RFdiffusion requires 16–32GB GPU memory and minutes per design Watson et al. [2023]. These requirements create accessibility concerns that the community must address through efficient architectures and shared infrastructure.

These technologies provide the computational substrate for design tasks. Section 5 examines the challenges limiting their practical applicability.

5 Challenges and Limitations

Despite remarkable progress, significant challenges limit practical applicability of AI-driven PPI design. This section examines major obstacles and outlines research directions.

5.1 Evaluation and Benchmarking

The evaluation crisis identified by Bernett et al. Tsishyn and Rooman [2024] represents a fundamental challenge: many reported high accuracies result from data leakage—train-test overlap at the sequence similarity level. When rigorous cross-validation prevents such leakage, model performance drops substantially. This suggests that many "successful" methods primarily learn to recognize sequence similarity rather than genuine interaction features.

Proposed Evaluation Standards. Based on this analysis, we propose standardized evaluation protocols:

1. *Rigorous Sequence Identity Thresholds*: Enforce maximum 25–30% sequence identity between train and test sets, verified through all-vs-all BLAST. However, even 30% thresholds may be insufficient: Tsishyn and Rooman Tsishyn and Rooman [2024] demonstrated that predictors show strong dataset dependence even with standard cross-validation, suggesting that sequence-level similarity alone does not capture all sources of bias. We recommend combining sequence thresholds with structural clustering (e.g., TM-score < 0.5) and ensuring no overlapping protein families between splits.

2. *Temporal Splits*: Evaluate on proteins discovered after training data cutoff date, preventing contamination through database annotations.

3. *Negative Sample Construction*: Use experimentally verified non-interactors (e.g., Negatome Blohm et al. [2014]) rather than random pairing, which introduces systematic bias.

4. *Multi-Dataset Validation*: Require consistent performance across independent benchmarks (SKEMPI Jankauskaite et al. [2019], DIP Salwinski et al. [2004], MINT Chatr-aryamontri et al. [2007]) to prevent overfitting to specific dataset characteristics.

5. *Negative Result Reporting*: Require reporting of failure rates and unsuccessful targets to combat publication bias. This faces implementation challenges: researchers have little incentive to publish negative results. We propose three mechanisms: (1) data repository requirements mandating deposition of all experimental results (successful and failed) as

a publication condition; (2) journal policy changes accepting "registered reports" where methodology is peer-reviewed before experiments begin; and (3) community recognition metrics rewarding rigor rather than just success rates. Methods achieving 20% success with complete transparency should be valued over methods claiming 50% success with selective reporting.

Design Evaluation Challenges. For design tasks, evaluation is more complex because ground truth does not exist. Success must be measured along multiple dimensions: success rate (fraction expressing, folding, binding), novelty (genuinely new vs. replicating known interactions), quality (affinities, specificities, stabilities), and practical utility (functioning in intended contexts). Reported success rates vary widely (5–50%) across studies, but direct comparison is difficult due to different experimental protocols and target difficulties.

5.2 Experimental Validation Bottleneck

Computational design outpaces experimental validation capacity. While methods generate thousands of candidates daily, validation remains time-consuming and expensive, creating a feedback bottleneck. Without systematic testing, methods cannot learn from failures Trepte et al. [2024]. Current practice validates only top-ranked designs, potentially missing high-performing designs that rank poorly computationally.

High-Throughput Solutions. Emerging platforms offer partial solutions: deep mutational scanning for parallel variant testing, yeast/phage display for library screening. However, these introduce biases—interactions functional on yeast surfaces may not work in mammalian cells. Cost structures create inequities: well-resourced labs validate more designs, accelerating their method development.

Infrastructure Needs. Addressing this bottleneck requires investment in: (1) centralized validation facilities accessible to the research community; (2) standardized protocols enabling cross-study comparison; (3) automated pipelines reducing human effort; and (4) data sharing platforms capturing both successes and failures. Estimated investment: $50–100M for shared infrastructure.

5.3 Exploring the Designable Space

A fundamental question remains unanswered: what fraction of theoretical sequence-structure space is actually designable? Not all sequences fold stably, and not all stable structures perform desired functions. The designable subset may be small Ingraham et al. [2023], and designed proteins with sequences unlike training data may fail despite computational predictions Dauparas et al. [2022].

Understanding designability boundaries has practical implications. If the designable space is limited, methods should focus on efficient exploration rather than random generation. If large but inefficiently sampled, improved search strategies would yield better designs. Systematic experimental characterization—deep mutational scanning combined with design—could map these boundaries, transforming design from trial-and-error to principled exploration.

5.4 Dynamic and Conditional Interactions

Most methods assume static structures and binary interaction states, ignoring biological reality: many PPIs are transient, condition-dependent, or allosterically regulated McGary et al. [2010]. Binding often involves conformational changes, induced fit, or conformational selection. Methods designing for single static structures may fail when actual interactions involve flexible regions or multiple conformations.

AlphaFold-based predictions capture some flexibility through ensembles but do not model binding kinetics or conformational dynamics Abramson et al. [2024]. Incorporating molecular

dynamics or ensemble approaches into design pipelines remains challenging. Designing proteins that bind only under specific conditions—activated by post-translational modifications or cofactors—requires modeling conditional dependencies that current methods largely ignore.

5.5 Current Approaches to Dynamic PPIs

Several computational approaches address dynamic PPIs, though none are yet integrated into generative design pipelines.

Molecular Dynamics-Enhanced Prediction. Enhanced sampling methods (metadynamics Barducci et al. [2008], replica exchange Sugita and Okamoto [1999]) can characterize conformational ensembles of protein interfaces. However, these simulations remain computationally expensive (10^3–10^5 GPU-hours per complex), limiting integration into iterative design workflows. Recent work combining AlphaFold with MD simulations Bezrodniy et al. [2024] demonstrated improved handling of flexible regions but increased computational cost 10-fold.

NMR Ensemble Integration. NMR structures capture conformational heterogeneity through ensemble refinement. Approaches like EnsembleDock Bolstad et al. [2003] dock against multiple conformers, selecting poses consistent across the ensemble. However, NMR structures represent only $\sim$5% of PDB entries, limiting applicability. For design, this suggests that generating multiple backbone conformations during diffusion and validating against all would improve robustness for flexible targets.

AlphaFold Confidence as Flexibility Indicators. AlphaFold's per-residue pLDDT scores correlate with experimentally observed flexibility—regions with pLDDT < 70 often correspond to intrinsically disordered regions Tunyasuvunakool et al. [2021]. Predicted aligned error (PAE) matrices can identify flexible inter-domain linkers. For design, this suggests: (1) targeting high-confidence (rigid) regions reduces dynamic complexity; (2) designing for conformational ensembles requires multiple backbone generation.

Conformational Selection vs. Induced Fit. The conformational selection model posits proteins sample multiple conformations, and binding partners select the most complementary one. Induced fit involves conformational change upon binding. Both operate in nature, but current design methods implicitly assume rigid-body recognition. A promising direction is multi-state design: generating binders against conformational ensembles and selecting candidates binding multiple states.

Limitations for Design. Current dynamic modeling faces three barriers: (1) computational cost (MD scales as $O(N^2)$ per timestep); (2) limited training data (few proteins have characterized dynamics); (3) evaluation challenges (measuring designed protein flexibility requires specialized experiments like hydrogen-deuterium exchange or single-molecule FRET). Near-term solutions include targeting rigid interfaces, using AlphaFold ensembles as flexibility proxies, and developing coarse-grained dynamics models.

5.6 Interpretability and Trust

Deep learning models underlying design methods are largely black boxes. When designs fail experimentally, current methods provide limited guidance for troubleshooting. Did failure result from misfolded backbone, poor sequence choice, or unpredicted interface geometry? Without interpretability, iterative design becomes trial-and-error.

For therapeutic applications, regulatory agencies require understanding mechanism-of-action. Black-box designs that cannot be explained or validated mechanistically may face regulatory hurdles Raybould et al. [2019]. Developing interpretable design methods—where decisions can be traced to physical principles—is essential for clinical translation. Near-term goal: design methods providing mechanistic explanations for interface affinity and specificity.

5.7 Ethical and Dual-Use Considerations

The ability to design novel proteins binding specific targets carries dual-use potential warranting careful consideration National Research Council [2004]. The same technologies could potentially design proteins with harmful properties, though several factors mitigate risks: technical barriers remain substantial, experimental validation creates bottlenecks, and synthesis services implement sequence screening.

The protein design community has embraced responsible practices: publishing methods openly while emphasizing beneficial applications, engaging with biosecurity experts, supporting screening frameworks. Continued vigilance and engagement with governance structures—including the Biological Weapons Convention, the National Science Advisory Board for Biosecurity (NSABB), and World Health Organization frameworks—will be essential as capabilities advance. Governance frameworks developed before AI-driven protein design may require updating to address democratization of design capabilities.

Governance Recommendations. We propose: (1) extending DNA synthesis screening protocols to cover AI-designed sequences; (2) establishing community standards for responsible publication of design methods; (3) engaging with regulatory bodies to develop updated frameworks balancing scientific openness with appropriate safeguards; and (4) developing anomaly detection systems for identifying potentially harmful design requests.

These challenges define the research frontier. Section 6 examines current applications despite these limitations.

6 Applications and Impact

The paradigm shift opens possibilities across therapeutic development, industrial biotechnology, and basic research (Table 3). This section examines major application areas.

Table 3: Selected applications of AI-designed PPIs

Domain	Application	Representative Results	Status
Therapeutics	SARS-CoV-2 binders	KD: 10–100 pM; 1000× potency Watson et al. [2023]	Preclinical
Therapeutics	PD-L1/VEGF-A binders	KD: 1–50 nM; 3–300× success Zambaldi et al. [2024]	Early-stage
Immunotherapy	IL-2 selectivity	10–100× selectivity improvement Fleming et al. [2021], Silva et al. [2020]	Preclinical
Enzymes	Cofactor binding pockets	Native-like geometry; catalytic activity Krishna et al. [2024]	Proof-of-concept
Research	Imaging probes	High-specificity intracellular labeling Zhang et al. [2024]	Established

6.1 Therapeutic Protein Design

Designed proteins can serve as therapeutics, drug components, or research tools. The global biologics market exceeded $350 billion in 2023, with AI-driven design offering reduced timelines and expanded target space Trepte et al. [2024].

Antibody and Binder Therapeutics. Traditional antibody development requires 12–18 months. AI-driven design accelerates this by enabling rapid candidate generation against new targets. For SARS-CoV-2, designed minibinders achieved picomolar affinities (KD: 10–100 pM) with improved stability—retaining function after weeks at 37°C versus days for many antibodies Watson et al. [2023]. The LCB1 minibinder showed neutralization potency (IC50: 15–50 pM) exceeding clinical-stage antibodies (IC50: 100–1000 pM). Unlike antibodies, minibinders are smaller (50–100 vs. 150 residues), more stable, and manufacturable in bacterial systems ($10–100/g vs. $100–1000/g in mammalian cells).

AlphaProteo designed binders against cancer targets (PD-L1, VEGF-A) with nanomolar affinities without experimental optimization Zambaldi et al. [2024]. Multiple companies (Generate Biomedicines, Insitro, Charm Bio) are developing AI-designed therapeutics, with clinical trials expected 2025–2026.

Cytokine Engineering. Natural cytokines often activate multiple receptor subtypes, causing off-target effects. Designed variants with improved selectivity could provide safer therapeutics. IL-2 variants demonstrating 10–100-fold improved selectivity between effector T cells and regulatory T cells could improve cancer immunotherapy while reducing toxicity Fleming et al. [2021], Silva et al. [2020]. Similar approaches apply to IL-4, IL-6, and interferon families.

Emerging Modalities. PROTACs (proteolysis-targeting chimeras) bring together target proteins and E3 ubiquitin ligases to trigger degradation. Over 20 PROTACs have entered clinical trials since 2019. Designing PPI interfaces between these components could improve efficiency. Bispecific antibodies—proteins binding two targets—offer immune cell redirection to tumors, with 10+ approved for oncology. Designing both interfaces while maintaining stability presents complex multi-objective optimization challenges.

6.2 Industrial and Economic Impact

Cost and Timeline Reduction. Traditional drug development costs exceed \$2.6 billion per approved drug, with biologics development requiring 12–18 months for lead identification Wouters et al. [2020]. AI-driven design reduces requirements by 10–100×: generating hundreds of high-quality candidates narrows experimental search dramatically. Traditional directed evolution screens 10^5–10^9 variants at \$1–5 million per optimized protein over 6–18 months. Faster development enables rapid response to emerging threats—demonstrated during COVID-19 when designed binders were generated within weeks of target availability.

Enzyme Engineering. Industrial biotechnology uses enzymes to produce chemicals, pharmaceuticals, and materials (market: \$10+ billion annually). Many transformations require multi-enzyme complexes channeling intermediates between active sites. RFdiffusionAA designed protein binding pockets for cofactors (heme, bilin) with native-like geometry and catalytic activity Krishna et al. [2024]. Extensions to multi-enzyme assemblies could enable metabolic pathway design for sustainable chemical production, biosynthesis of complex natural products, and bioremediation.

Agricultural Applications. Designed binding proteins could protect crops from pathogens by interfering with virulence factors, improve nitrogen fixation by engineering Rhizobium-plant symbiosis, or enable biosensing of environmental contaminants.

6.3 Research Tools and Basic Science

Mechanism Validation. Computational predictions can be tested through designed proteins. If a model predicts specific interface residues mediate binding, mutation in a designed binder should disrupt interaction. This enables systematic dissection of interaction mechanisms beyond natural variation Watson et al. [2023].

Imaging and Detection. Fluorescent binders enable live-cell visualization of protein localization and dynamics Zhang et al. [2024]. Designed nanobodies fused to fluorescent proteins provide intracellular imaging capabilities impossible with traditional antibodies. This combination enables powerful imaging applications.

Synthetic Biology. Engineered signaling pathways could reprogram cellular behavior. The synNotch system combines sensing domains with transcriptional responses Morsut et al. [2016]. Chen et al. demonstrated design of protein-interaction specificity from sequence-to-sequence learning, enabling precise control over which proteins interact Chen et al. [2019b]. Applications include therapeutic cells sensing disease markers, industrial microbes optimizing metabolism, and biosensors detecting environmental conditions.

Structure-Function Exploration. Systematic exploration requires large variant sets. High-throughput characterization of hundreds of designs builds datasets revealing design principles. Deep mutational scanning combined with design maps fitness landscapes, identifying prediction-experiment discrepancies that point to understanding gaps Goldenzweig et al. [2016].

6.4 Economic and Accessibility Considerations

Computational Divide. Training large PLMs requires hundreds of GPU-years. Inference costs vary: ProteinMPNN runs CPU-capable in milliseconds; RFdiffusion requires 16–32GB GPU for minutes per design Watson et al. [2023]. AlphaFold3 access is limited to web interfaces with usage caps. This creates inequities: well-resourced institutions can run thousands of designs while resource-limited labs cannot compete.

Democratization Strategies. Open-source alternatives (RFdiffusion, ProteinMPNN, ESMFold) help but require expertise and infrastructure. Cloud computing provides partial solutions but introduces ongoing costs. Addressing accessibility requires: (1) maintaining open implementations; (2) providing cloud access for resource-limited researchers; (3) developing efficient architectures reducing computational requirements; and (4) establishing shared infrastructure.

Environmental Impact. Training large models emits substantial carbon. As the field scales, sustainable practices—efficient architectures, shared pretrained models, optimized inference—become increasingly important. Estimated carbon footprint: single large PLM training equals several transatlantic flights; thousands of design iterations compounds this impact.

These applications leverage the ability to create proteins rather than search for natural variants. As methods mature, scope will expand from individual binders to multi-protein systems and programmable biological functions. Realizing this potential requires addressing challenges outlined in Section 5.

7 Future Perspectives

The paradigm shift represents not an endpoint but a transition toward deeper AI-biology integration. This section outlines emerging directions.

7.1 Prediction-Design Integration

Current workflows treat prediction and design as separate stages. Future systems will optimize this loop end-to-end through active learning—prioritizing experiments that maximally reduce uncertainty about design rules, with each result updating the model Dallago et al. [2024]. Unified models could predict interaction properties, generate designs, and evaluate quality within consistent frameworks, enabling more efficient design space exploration.

7.2 Toward Protein Intelligence

Systems operate at different capability levels: understanding (parsing structures and functions), prediction (inferring interactions), and design (creating proteins with desired functions). The emerging fourth level—**creation**—would generate biological systems solving problems evolution never addressed, requiring reasoning about multiple proteins within functional systems. As language models scale, few-shot and zero-shot design capabilities may expand, enabling rapid adaptation to new challenges without task-specific training Lin et al. [2023].

7.3 Methodological Advances

Dynamic Modeling. Current methods design static interfaces, but many functional PPIs involve conformational changes and allosteric regulation. Integrating molecular dynamics with generative design could enable proteins with dynamic binding behavior. Key milestone: designing an allosterically regulated binder switching affinity in response to a small molecule or post-translational modification.

Multi-Component Systems. Extending from pairwise interactions to 3+ protein complexes would enable signaling pathways, metabolic channels, and synthetic cellular machines.

This requires solving combinatorial explosion of interface configurations while maintaining assembly stability. Target capability: designing a 5-protein complex with specified architecture and function.

Enhanced Evaluation. Addressing the evaluation crisis requires: (1) community-agreed benchmark datasets with standardized splits; (2) negative result repositories capturing failures; (3) prospective evaluation requirements for published claims; and (4) computational validation standards (multiple prediction methods, confidence thresholds, structural quality metrics).

7.4 Experimental Infrastructure

Computational design outpaces validation by 100–1000×. High-throughput platforms testing thousands of designs weekly with standardized protocols would close this gap. Required investment: $50–100M for shared infrastructure. Emerging solutions include deep mutational scanning for parallel variant testing, and yeast/phage display for library screening, though these introduce system-specific biases.

7.5 Interpretability and Clinical Translation

For therapeutic applications, regulatory approval requires understanding *why* a design works. Black-box methods face hurdles without mechanistic explanations Raybould et al. [2019]. Interpretable design methods—where decisions trace to physical principles—are essential. Near-term goal: methods providing mechanistic explanations for interface affinity and specificity.

Regulatory frameworks are evolving to address AI-designed therapeutics. The U.S. FDA's guidance on AI/ML-enabled medical devices emphasizes "predetermined change control plans" for continuously learning systems. For AI-designed proteins, this suggests three requirements: (1) mechanistic models explaining binding predictions; (2) validated surrogates for clinical endpoints; and (3) robust uncertainty quantification. The European Medicines Agency (EMA) has similarly highlighted the need for explainable AI in drug development. These regulatory expectations will shape methodological priorities for the field.

7.6 Paradigm Shifts in Research Practice

From Discovery to Engineering. Traditional biology is discovery-driven; design-driven biology inverts this—defining desired functions, generating designs, validating experimentally Bennett et al. [2024]. This engineering mindset treats biology as design space to be explored.

From Trial-and-Error to Rational Design. Drug discovery historically relied on high-throughput screening. AI-driven design enables targeted generation, reducing screening scale while improving candidate quality Zambaldi et al. [2024].

From Observing to Creating. Most fundamentally, the design paradigm enables creating what evolution has not. Designed proteins with sequences unlike any natural protein test whether understanding generalizes beyond evolutionary constraints Ingraham et al. [2023].

7.7 Accessibility and Equity

Current computational requirements create inequities. Addressing these requires: (1) efficient architectures reducing GPU requirements; (2) cloud-based access for resource-limited researchers; (3) shared computational infrastructure; and (4) open-source implementations maintained long-term. Environmental sustainability concerns also grow with scale—efficient methods and shared infrastructure reduce collective carbon footprint.

7.8 Grand Challenges for the Next Five Years

Challenge 1: Designability Mapping. What fraction of sequence-structure space is designable? Systematic experimental characterization through large-scale design and testing would transform design from trial-and-error to principled exploration. Timeline: 3–5 years with coordinated community effort.

Challenge 2: Dynamic Interaction Design. Designing proteins with conformational flexibility or conditional binding requires integrating dynamics with generation. Milestone: designing a switchable binder with 10× affinity change upon ligand binding.

Challenge 3: Validation Infrastructure. Closing the computational-experimental gap requires shared high-throughput facilities. Investment: $50–100M for infrastructure accessible to the research community.

Challenge 4: Interpretability. Clinical translation requires explainable designs. Goal: methods providing atomic-level explanations linking design decisions to predicted properties.

Challenge 5: Standards and Governance. Updated frameworks for responsible development, including synthesis screening protocols and publication standards.

The prediction-to-design transition marks a shift from AI as passive observer to AI as active creator. Whether this transition fulfills its promise will be determined by collective efforts in the coming years. The technical foundations are in place; the challenges are known; the opportunity is unprecedented. The conclusion synthesizes these themes and offers final perspectives.

8 Conclusion

Computational protein-protein interaction research is undergoing a fundamental transformation. This review has traced the trajectory from prediction—inferring whether, where, and how proteins interact—to design—creating proteins with desired interaction properties. This paradigm shift represents a qualitative change in how AI interfaces with biological discovery.

The technological foundations are established. Protein language models encode evolutionary and structural information at scale Rives et al. [2021], Lin et al. [2023]. Diffusion models generate novel structures conditioned on functional constraints Watson et al. [2023], Ingraham et al. [2023]. Equivariant neural networks preserve geometric symmetries essential for accurate modeling Jing et al. [2021], Fuchs et al. [2020]. Together, these enable design capabilities implausible just five years ago.

Yet significant challenges remain. The evaluation crisis suggests reported accuracies may be inflated Tsishyn and Rooman [2024]. Success rates vary widely (10–50% across targets), with negative results systematically underreported. Experimental validation limits iterative improvement. Dynamic interactions, conditional specificity, and multi-protein complexes remain largely beyond current capabilities.

The implications extend beyond technical advances. Drug discovery transitions from high-throughput screening to targeted generation Zambaldi et al. [2024]. Synthetic biology can design interaction networks rather than repurposing natural components. Basic research gains hypothesis-driven experimental tools.

Perhaps most significantly, this transition reflects a philosophical shift: from AI as a tool for understanding what nature produced to AI as a partner in creating what nature has not. This does not diminish prediction—prediction remains the foundation upon which design builds Jumper et al. [2021], Dauparas et al. [2022]. Rather, it positions prediction as capability to be leveraged rather than endpoint in itself.

The coming years will determine whether current successes represent a sustained revolution or incremental advance peak. Progress depends on: better evaluation, faster validation, deeper understanding of designability, dynamics modeling, and interpretable methods. If addressed, the prediction-to-design transition will likely be remembered as a pivotal moment when

computational biology substantially expanded its creative capabilities.

Key Points:

1. AI-driven PPI research is transitioning from prediction to design—a qualitative expansion of capabilities enabled by protein language models, generative diffusion models, and equivariant neural networks.
2. This constitutes a paradigm shift in the Kuhnian sense: reconceptualizing what questions are meaningful and what methods are appropriate.
3. Design capabilities transform drug discovery, synthetic biology, and basic research, but success rates vary dramatically by target (10–50%).
4. Critical challenges—evaluation bias, validation bottlenecks, interpretability gaps, ethical considerations—must be addressed to realize the paradigm's potential.
5. The future lies in prediction-design integration: AI systems iteratively creating, validating, and refining in closed-loop workflows.

Implications:
For Researchers: Develop rigorous evaluation protocols, publish negative results, build computational-experimental feedback loops. Highest-impact work addresses where current methods fail: dynamic interactions, conditional specificity, multi-component assemblies.
For Industry: Prepare for accelerated discovery timelines and expanded druggable space. Investment in experimental validation infrastructure will differentiate successful programs. Economic advantage shifts from largest compound libraries to most efficient design-test cycles.
For Funders: Support fundamental challenges—better energy functions, dynamics modeling, interpretable design—rather than incremental benchmark improvements. Infrastructure for validation and data sharing amplifies methodological advances.

The prediction-to-design paradigm shift marks a transition from AI as passive observer of biology to AI as active creator of biological function. The technical foundations are in place; the challenges are known; the opportunity is unprecedented. Whether this transition fulfills its transformative promise will be determined by the collective efforts of researchers, developers, and practitioners in the coming years.

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